



XMBA 200S: Data and Decisions

Chapter 14: Sampling Variation and Quality

Instructor: Reed Walker

Chapter 13: Recap - Eliminating Bias

The best way to get a representative sample (i.e. eliminate bias) is to **pick members of the population at random**

If you select at random, by design **any characteristic** in the population will be not correlated to being included in the sample (say, driving style of the owner in VDS)

- This includes things you do not see (unobservables)
- SRS ensures that on average a sample mimics the population

How Do We Assess Likelihood of Bias?

I downloaded a database of used car sales from Kaggle.com

The table below describes this population (price in 10s)

Do samples A, B, and C *look like* random samples from this **sampling frame**? Let's compare means

```
. sum selling_price km_driven year
```

Variable	Obs	Mean	Std. Dev.	Min	Max
selling_price	8,128	638.2718	806.2534	29.999	10000
km_driven	8,128	69819.51	56550.55	1	2360457
year	8,128	2013.804	4.044249	1983	2020

```
. sum selling_price km_driven year
```

Variable	Obs	Mean	Std. Dev.	Min	Max
selling_price	8,128	638.2718	806.2534	29.999	10000
km_driven	8,128	69819.51	56550.55	1	2360457
year	8,128	2013.804	4.044249	1983	2020

Sample A

Variable	Obs	Mean	Std. Dev.	Min	Max
selling_price	3,942	1036.67	1009.499	451	10000
km_driven	3,942	57805.41	62602.07	1000	2360457
year	3,942	2016.192	2.273285	2005	2020

```
. sum selling_price km_driven year
```

Variable	Obs	Mean	Std. Dev.	Min	Max
selling_price	8,128	638.2718	806.2534	29.999	10000
km_driven	8,128	69819.51	56550.55	1	2360457
year	8,128	2013.804	4.044249	1983	2020

Sample B

Variable	Obs	Mean	Std. Dev.	Min	Max
selling_price	1,050	1870.828	1640.533	75	10000
km_driven	1,050	40276.09	35358.56	1000	230000
year	1,050	2016.373	3.102269	1997	2020

. sum selling_price km_driven year

Variable	Obs	Mean	Std. Dev.	Min	Max
selling_price	8,128	638.2718	806.2534	29.999	10000
km_driven	8,128	69819.51	56550.55	1	2360457
year	8,128	2013.804	4.044249	1983	2020

Sample C

Variable	Obs	Mean	Std. Dev.	Min	Max
selling_price	1,107	653.9824	820.3079	31	6223
km_driven	1,107	68072.81	48304.86	1000	577414
year	1,107	2013.828	4.050025	1997	2020

Main Issues in Data Production: Forms of Selection Bias

1. Sampling Frame Bias
2. Self-Selection Bias/Voluntary Response Bias
3. Survivorship Bias
4. Measurement Bias

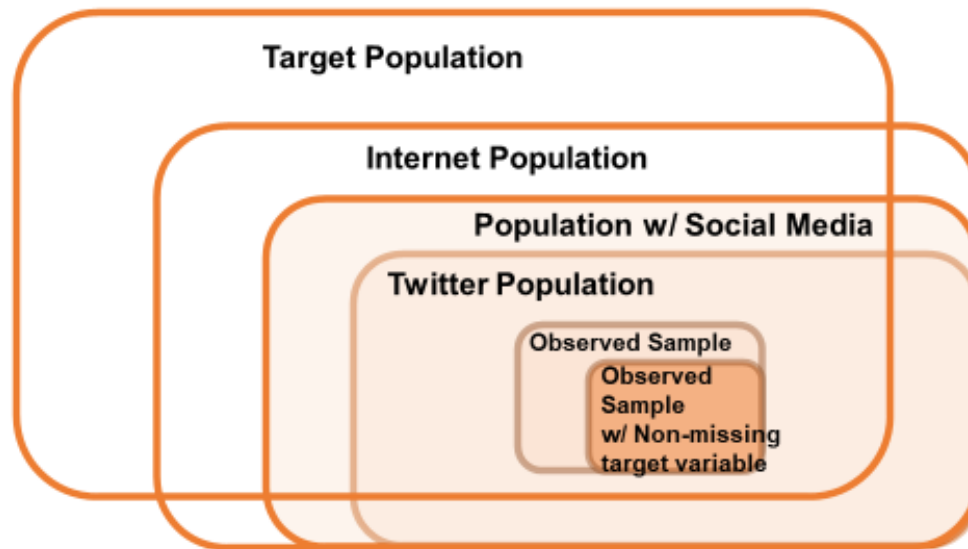
1. Sampling Frame Bias

Population of Interest: 20-35 Year Old Men

Survey Frame: 20-35 year old Twitter users that share age/gender in profile



Coverage & Selectivity in Twitter Data

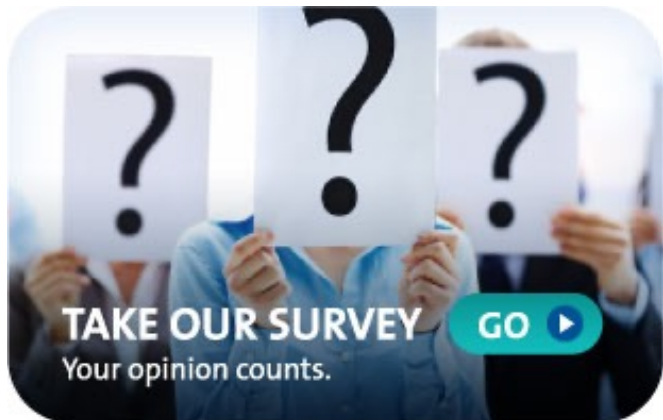


Source: Eurostat

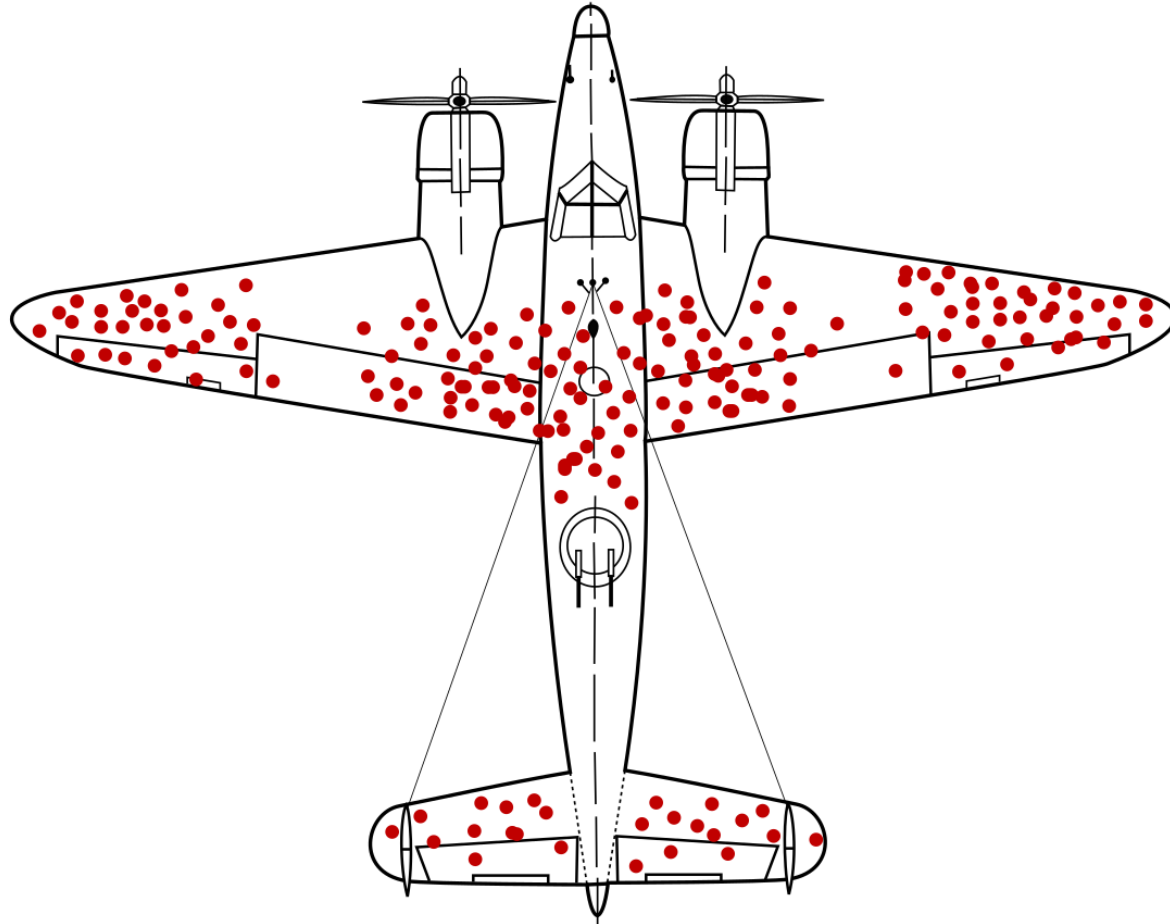
2. Self-Selection Bias

Voluntary Response Sample:

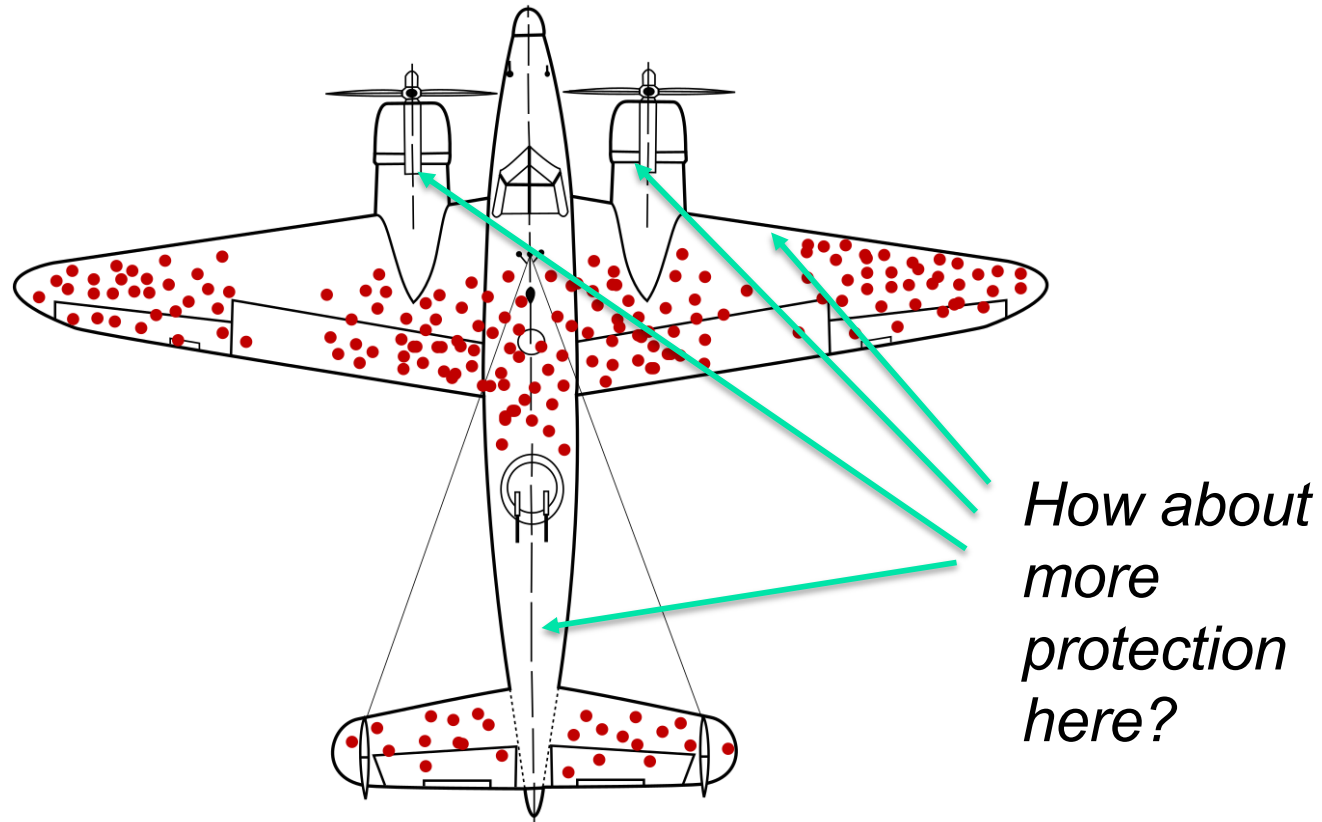
- A sample consisting only of individuals who volunteer to participate in a survey
- The big challenge with these surveys is that they are not representative
- E.g. Non Random Non Response of likely voters



3. Survivorship Bias: Example



3. Survivorship Bias: Example

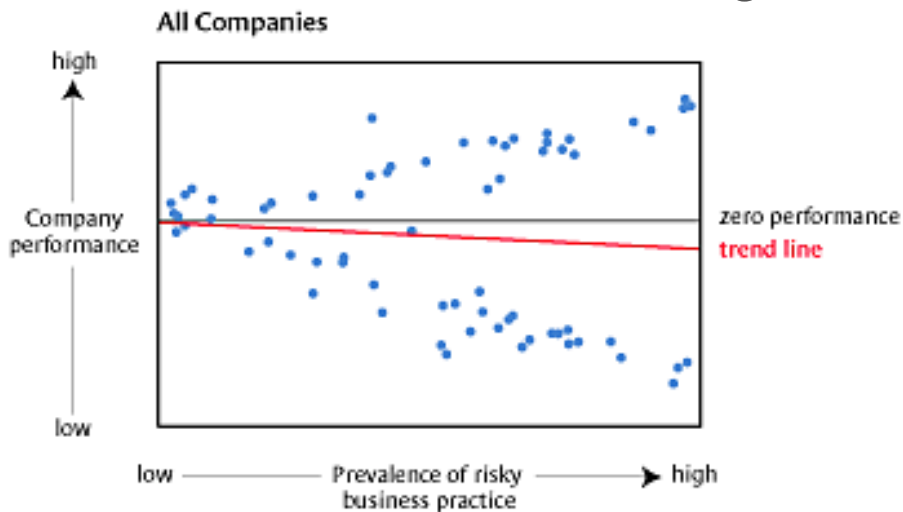


Basing your analysis only on the subset of your original sample that clears a certain hurdle or remains in a sample produces bias

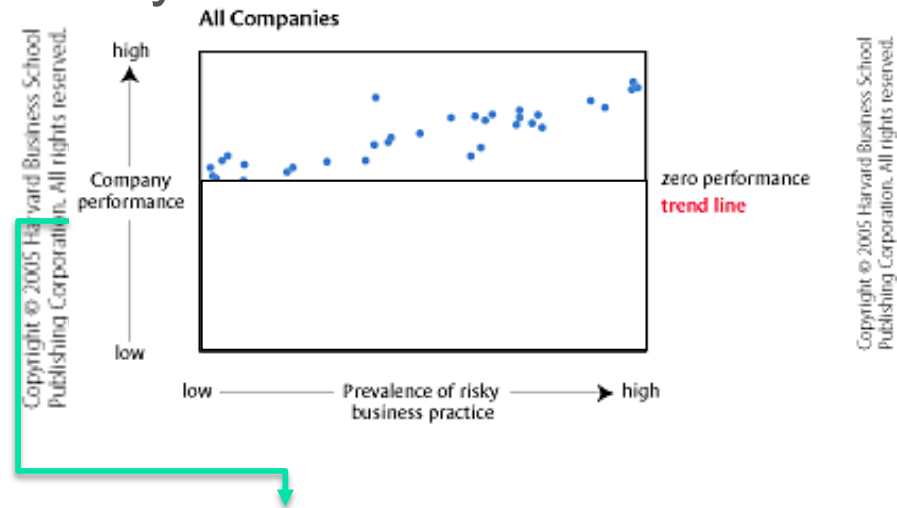
- Think of comparing firms implementing a certain risky business practice or not. You can only observe firms that stay in business.
- Imperfect Solutions: “Selection Correction” https://en.wikipedia.org/wiki/Heckman_correction

3. Survivorship Bias: Example

Reality: Risky mgmt. practice produces either big gains or losses & it's bad on average



Observed: Only firms that got lucky and obtained big gains stay viable



But what you'd conclude (wrongly):
That risky mgmt. practice produces big gains!

4. Measurement Bias

Bias coming from measurement. It comes in many forms and for many reasons

- Experimenter demand effect: people respond to a survey in a way that tries to please the surveyor. “Are you likely to vote?”
- Strategic responses: It may also affect how firms answer official government surveys (e.g. BLS, FED, Census). Firms may lie, wishing to avoid being audited, or to shade up regulatory expenditures, etc.

Samples with Selection: Take Aways

The key questions in data production:

1. Who is left in my sample?
2. Who is left out?

If the two sets are

- Different → You are in trouble
- The same → You are ok

Remember: **this is why random samples are so valuable**

- Those left in are in no way systematically different from those left out. SRS, when done on the correct frame, are representative!

Mitigation #1: Professional Consumer Panels



INSIGHTS SOLUTIONS NEWS CENTER ABOUT

SOLUTIONS

CONSUMER PANELS

Measuring what consumers buy is at the core of Nielsen. We track consumer behavior for more than 250,000 households in 25 countries through our industry-leading consumer panel. Our consumer panel is robust, trusted, credible and one of the biggest panels per capita in the world.

Mitigation #2: Recruitment Techniques

The Tilt

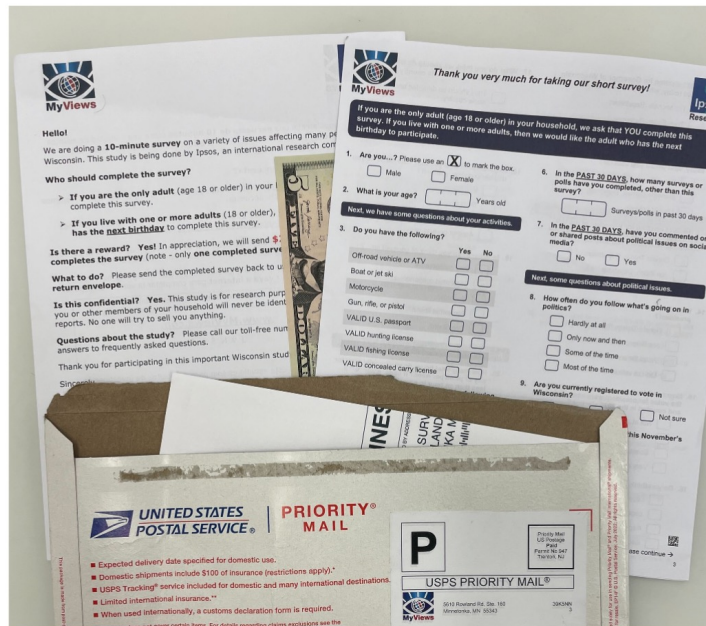
Are the Polls Still Missing ‘Hidden’ Republicans? Here’s What We’re Doing to Find Out.

There’s no breakthrough in our experiment yet, but the differences are intriguing.



By Nate Cohn

Nov. 8, 2022



A copy of the survey sent to households across Wisconsin.

How Voters Reached by Mail — and Paid \$25 — Are Different From Those Reached by Phone

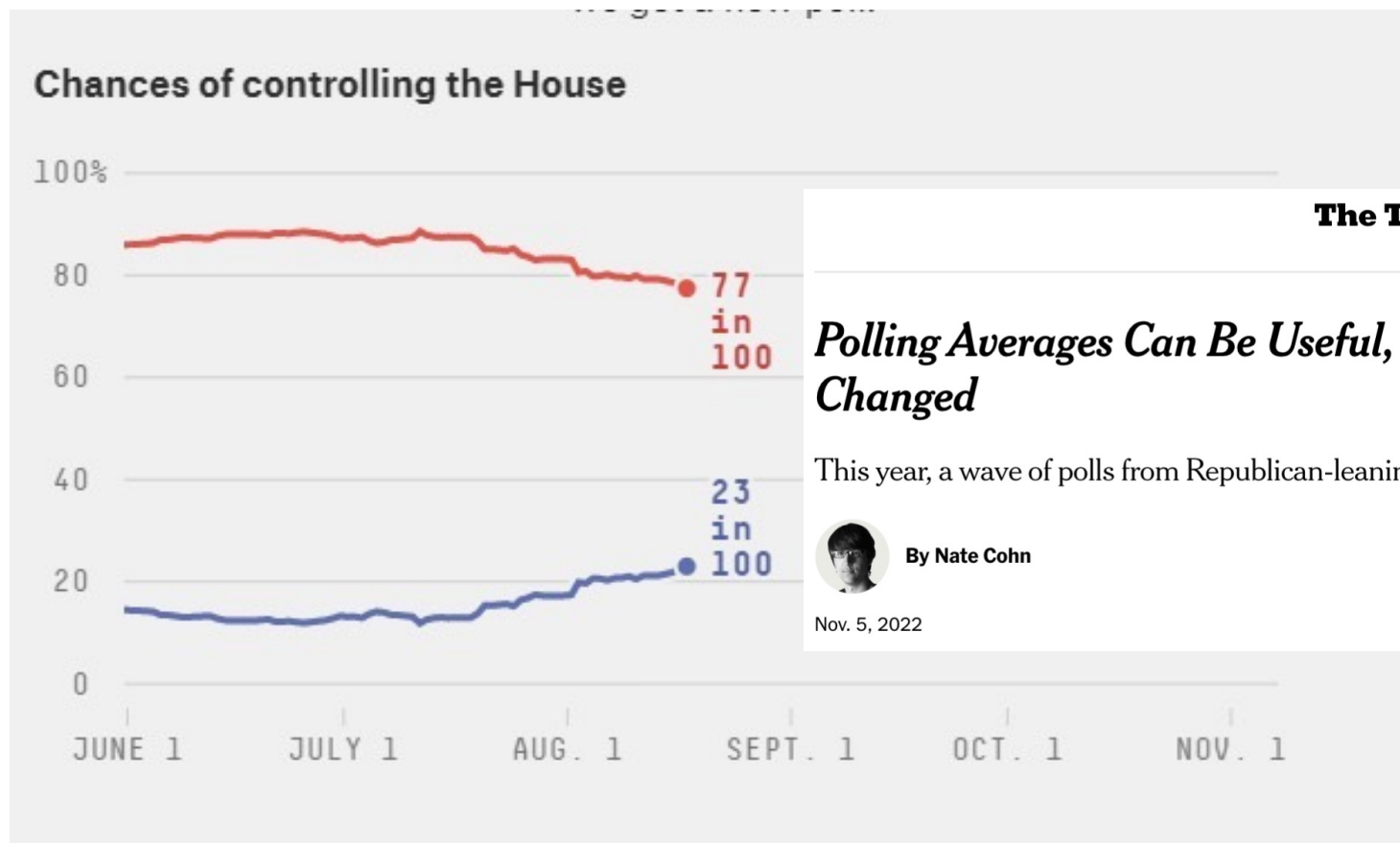
While respondents who received \$25 were significantly more likely to participate in a political survey, they were not meaningfully different in their political views or composition than a similar group of voters who did not receive money.

These voters have similar political preferences

	IPSOS MAIL	NYT/SIENA PHONE
Senate		
Ron Johnson Republican	44%	43%
Mandela Barnes Democrat	44%	45%
Governor		
Tim Michels Republican	42%	41%
Tony Evers Democrat	46%	46%

See Lecture 1 bCourses “Articles”

Mitigation #3: Combination of Polls to Average out Bias



The Tilt

Polling Averages Can Be Useful, but What's Underneath Has Changed

This year, a wave of polls from Republican-leaning firms is driving the averages.



By Nate Cohn

Nov. 5, 2022

Not straightforward and not a panacea.
See Lecture 1 bCourses "Articles"

Mitigation #4: Bias Correction Methods / Reweighting Data

Sometimes there is nothing you can do to prevent selection/sampling bias

If you understand the direction in which representativeness fails, you can correct with reweighting techniques

- i.e. groups that are oversampled can be down weighted or vice versa

Where Does Data Come From?

Weight and weighting



Emily Oster ✓
Dec 5

♡ 22

Most common technique but not perfect. See Lecture 1 bCourses “Articles”
+ example in Lecture 2 bCourses.

Mitigation #5: Worst Case Bounds

If you know the response rate, you can sometimes “impute” the missing observations using cases or “bounds”.

Example: political polling

true support =
(share who respond) $\times$ (support among respondents) +
(share who don't) $\times$ (support among nonrespondents)

You don't know the red, but you can assume different cases:

- Case 1: everyone who didn't respond supports, or
- Case 2: everyone who didn't respond opposes.

These cases bracket the true support

Most credible but often not informative. See example in Lecture 2 bCourses.

Chapter 14, Data and Decisions Part 1:

The data you want versus the data you've got

- Managers (and others) almost never have all of the data they want...
 - On their customers
 - On their production facilities
 - On their employees' productivity
- If you can only collect **a sample of data**
 - **This class:** What can you say about what is generally going on?

The set up for today

The fundamental problem is you need to make decisions observing limited data

You'll choose to act when information is “compelling”

But what does compelling mean?

- When to buy a stock?
- When to promote a worker?
- When to stop a production process?

What we'll discuss today

Why the tables in the back of the textbook are so useful...

- We can put a **measure** on how surprising a particular result is

In many settings, this will help you make decisions

- e.g., an observation is far enough from a **benchmark**, you will act

Control limits are thresholds that you set that determine when you act

- How to set these limits? Balance risks of two types of errors

Quality Control Thru Testing

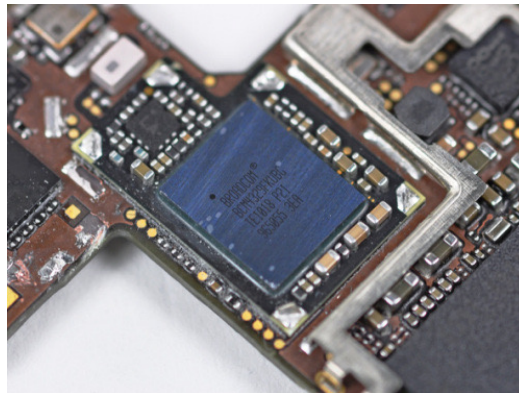


Quality Control Thru Testing



Sampling Distribution of the Mean

- A manufacturer of GPS chips selects samples for highly accelerated life testing (HALT).
- How should managers monitor these tests to ensure proper functioning of the production process?



Sampling Distribution of the Mean

Random Variation or Change in Process?

- There are 16 total tests and the score is the number of tests passed (i.e. an integer from 0-16)
- Even when functioning properly there is variation among HALT scores (recorded as number of tests passed)
- When things are functioning properly, on average chips should pass $\mu = 7$ tests with a standard deviation $\sigma = 4$
- When do you stop the assembly line?

Two Types of Errors

Type I Error: This is the mistake of taking action when no action is needed.

Type II Error: This is the mistake of failing to take action when action is needed.

Each type of error has a cost

- Make decisions trading off these costs

Type I Errors Are Costly

The New York Times | <http://nyti.ms/1W3vxR0>

HEALTH

American Cancer Society, in a Shift, Recommends Fewer Mammograms

By **DENISE GRADY** OCT. 20, 2015

One of the most respected and influential groups in the continuing breast-cancer screening debate said on Tuesday that women should begin mammograms later and have them less frequently than it had long advocated.

YOUR HEALTH

If you're 40, it's time to start mammograms, according to new guidelines

May 9, 2023 • 11:00 AM ET

Heard on [All Things Considered](#)



Allison Aubrey

See Lecture 2 bCourses “Articles”

Type II Errors Are Costly



Alarms, detectors disabled so top rig officials could sleep

The Deepwater Horizon's fire and gas leak alarms were disabled for at least a year to prevent false alarms from waking up rig leaders, a chief engineer told federal investigators.

Other Type 1 and Type 2 Error Tradeoffs

Examples:

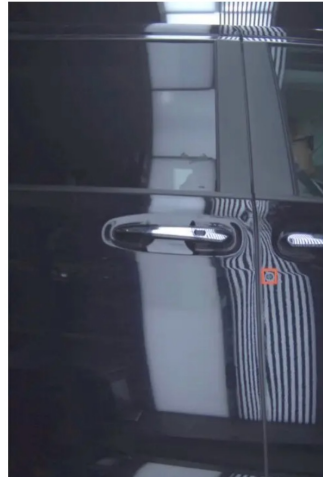
- Credit card fraud
- Technology and car rentals



‘It Could Have Been a Shadow’



An image captured by the scanner showed the passenger-side front door when Kelly Rogers picked up her rented minivan in Atlanta. Kelly Rogers

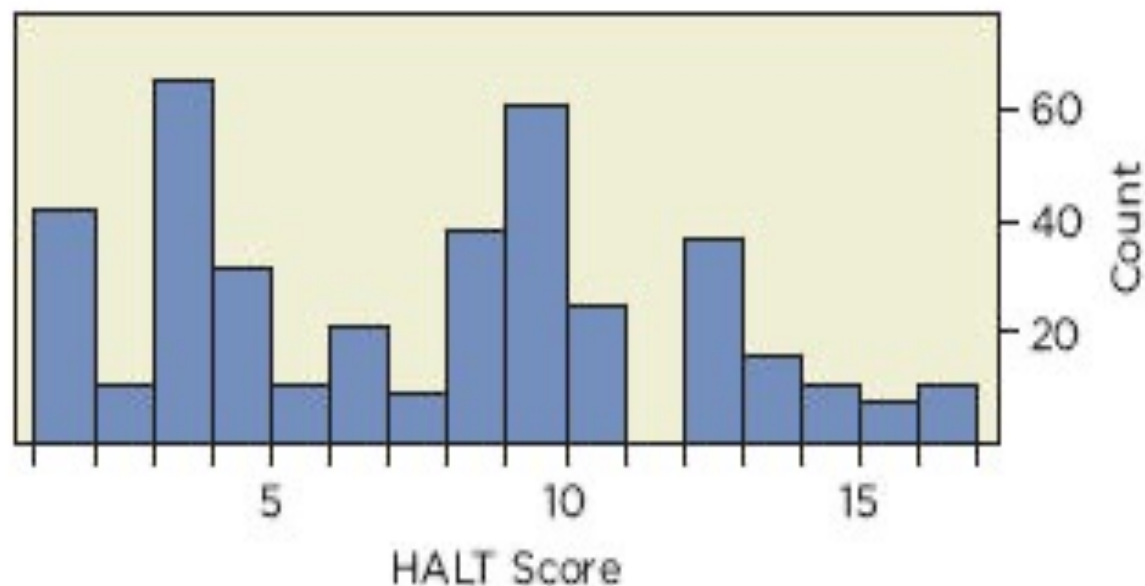


The scanner captured the door of Ms. Rogers's rented minivan when she was returning it, showing the spot Hertz claimed was a dent. Kelly Rogers

A.I. Is Making Sure You Pay for That Ding on Your Rental Car

Hertz and other agencies are increasingly relying on scanners that use high-res imaging and A.I. to flag even tiny blemishes, and customers aren't happy.

Distribution of Individual HALT Scores from Day 1

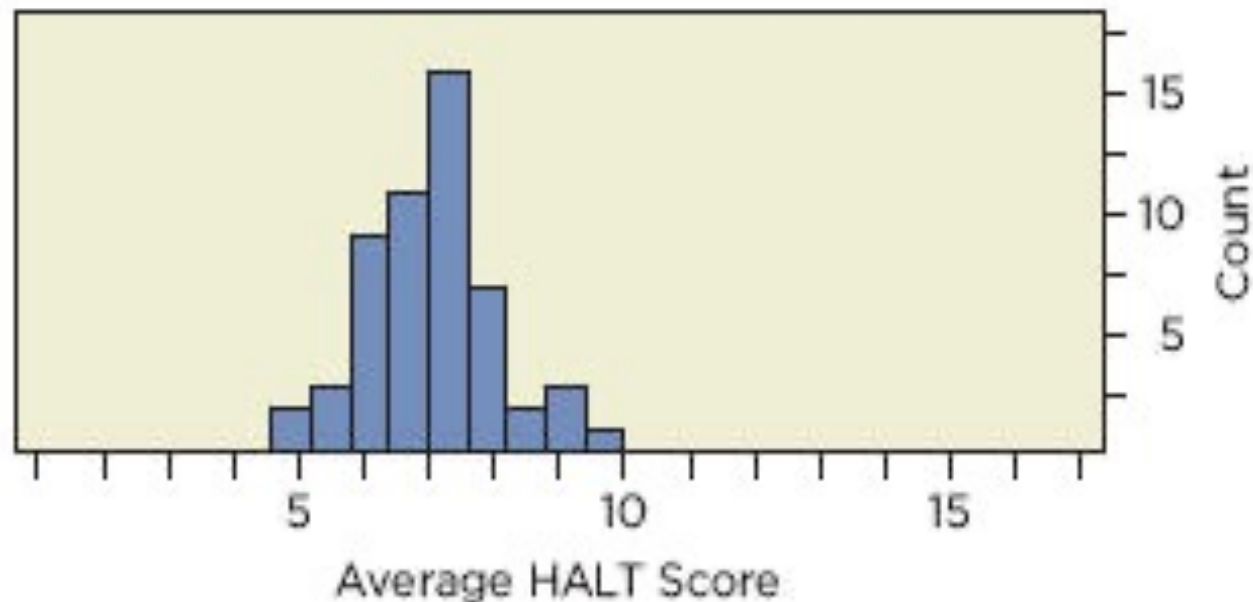


We destroyed our first 1000 chips to learn about population μ (mean) and σ (standard deviation) HALT scores in production

- Next: we will use this information to monitor ongoing production

Distribution of 50 Mean HALT Scores

Each mean or average is constructed from 20 chips, $n=20$

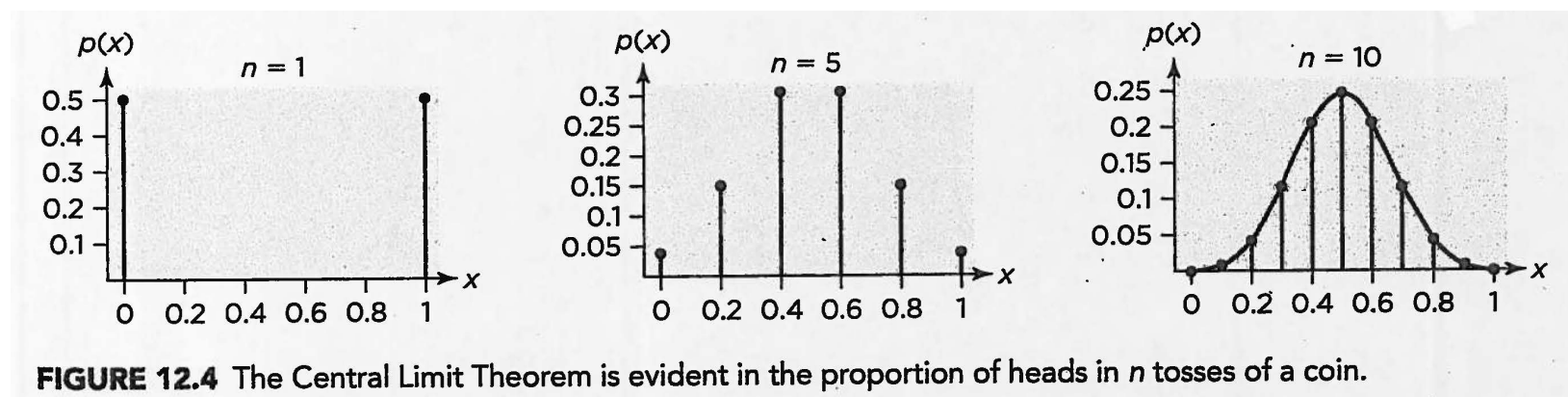


Distribution of 50 sample means from the next 50 days
(i.e. 50 different samples of $n=20$ chips, one sample per day)

Benefits of Averaging

The distribution of the sample mean is approximately normal, *regardless* of the distribution of the underlying data (Central Limit Theorem).

- This is very useful, as we can take any weird looking dataset, and if we look at a mean/average from this dataset, we know something about how noisy that statistic is.



Sampling Distribution of the Mean

A sample mean is a random variable (“sampling variability”)

Sample means are normally distributed if either:

- The individual values are normally distributed.

OR

- The sample size condition is satisfied (i.e. because of the Central Limit Theorem).

When can you assume normality?

Sample Size Condition:

Heuristic from Textbook: if the sample size n is larger than 10 times the absolute value of the kurtosis (K_4).

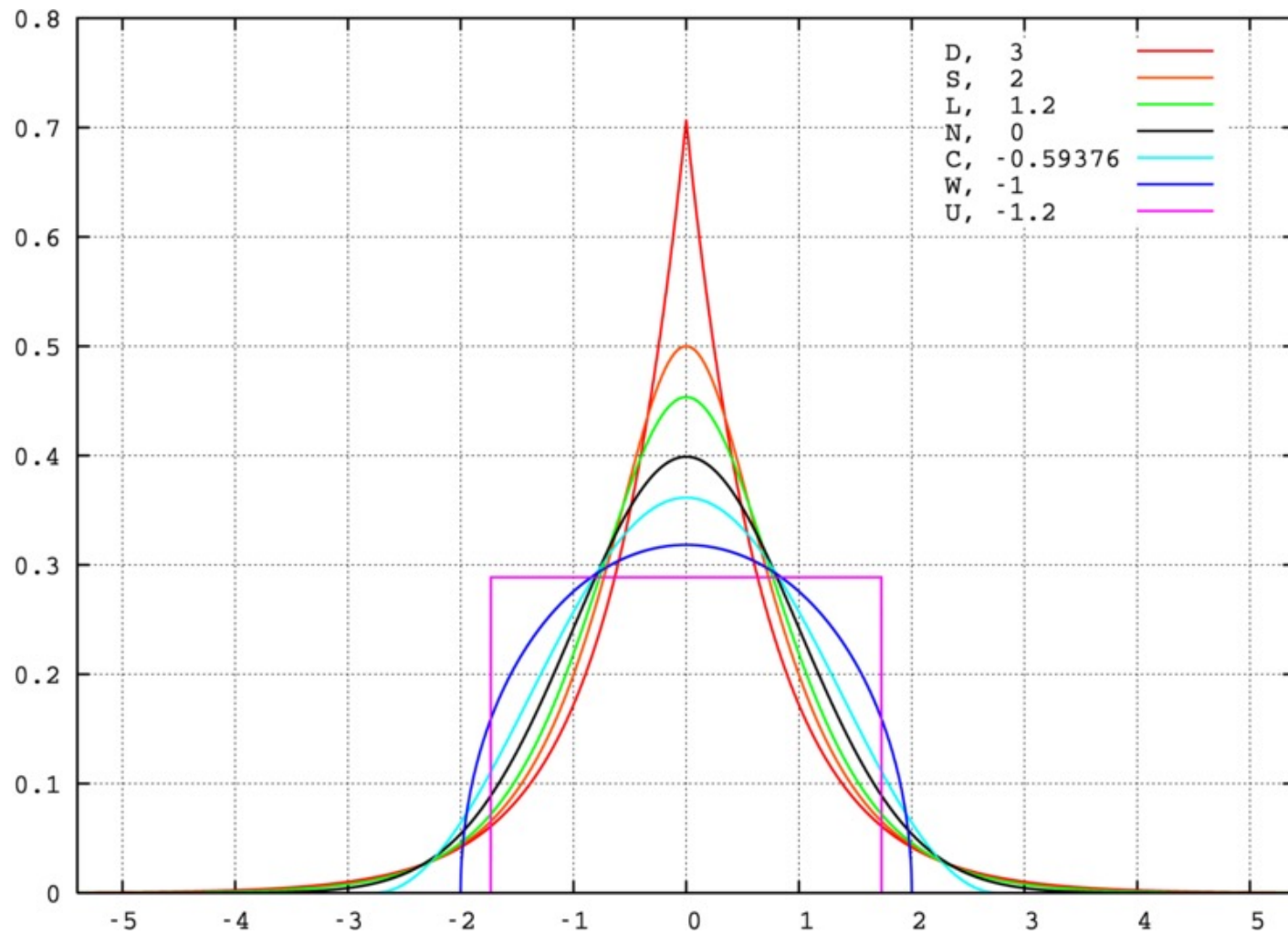
$$n > 10 |K_4|$$

then the distribution of the sample mean can be approximated by a normal distribution

Kurtosis: Measures the prevalence of outliers.

- $K_4 = 0$ for normal data.

Kurtosis



Sampling Distribution of the Mean

Definition: Standard Error of the Mean

- A measure of the sample-to-sample variability in sample mean (i.e. how noisy is our statistic)
- Calculated as $\sigma/\sqrt{n}$

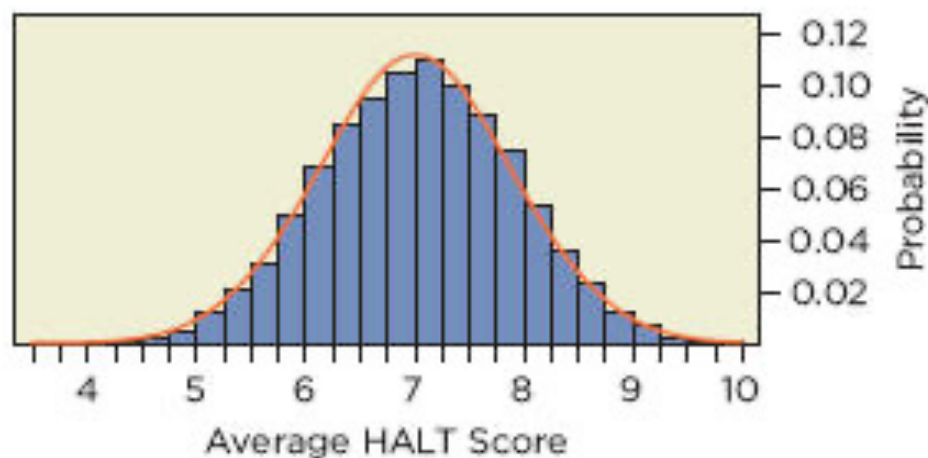
Standard Error of the Mean

- The standard error is proportional to σ .
 - When the population you are sampling is more variable (higher σ), the sample mean is more variable.
- The standard error is inversely proportional to the $\sqrt{n}$
 - The sample mean is less variable with larger samples
- The standard error does not depend on the size of the population.

Sampling Distribution of the Mean

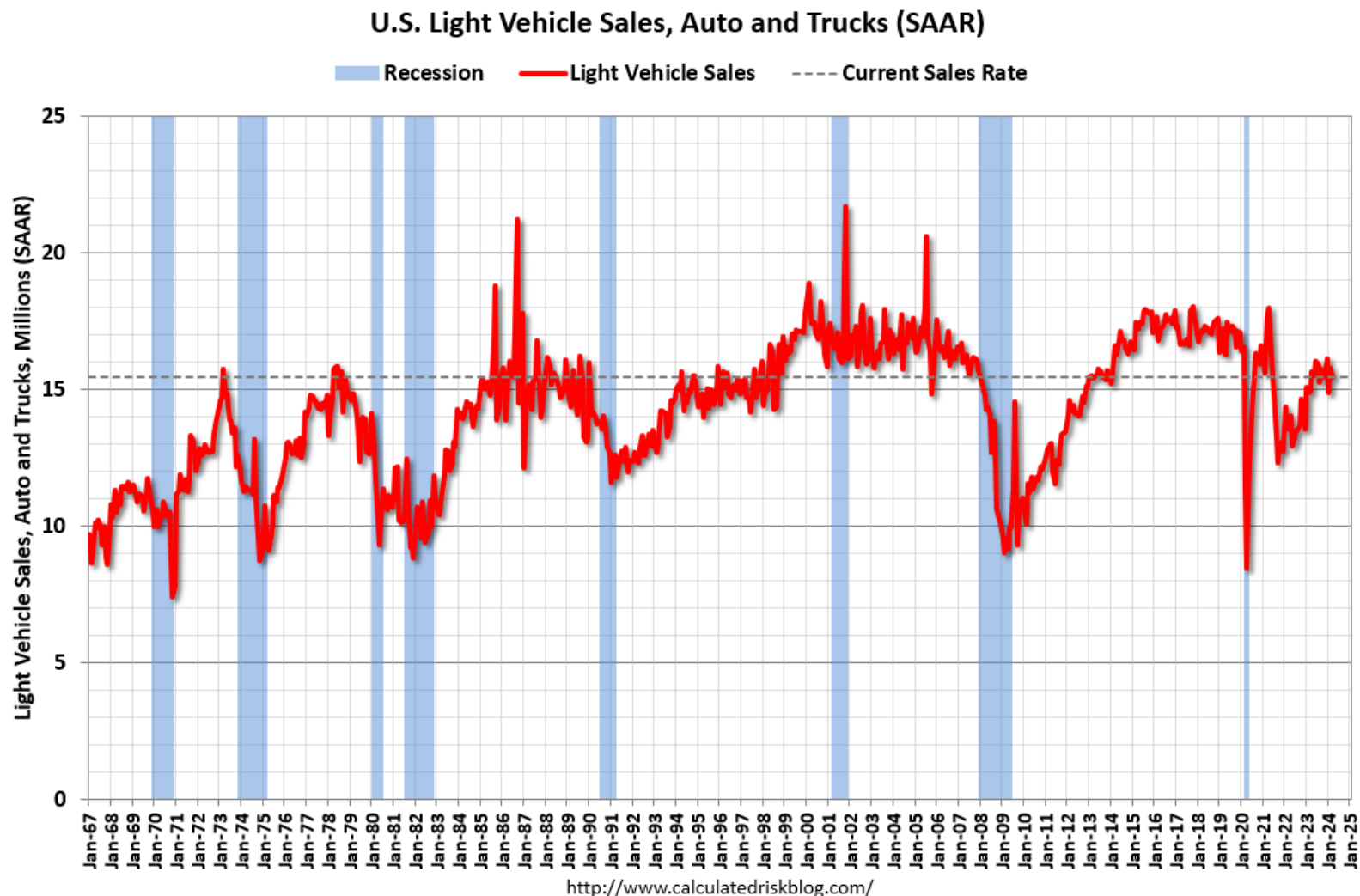
Sampling Distribution for Mean HALT score

$$\bar{X} \sim N\left(\mu = 7, \frac{\sigma^2}{n} = \frac{16}{20} = 0.8\right)$$



Can you translate the notation into English?

Do Large Populations Require Large Samples?



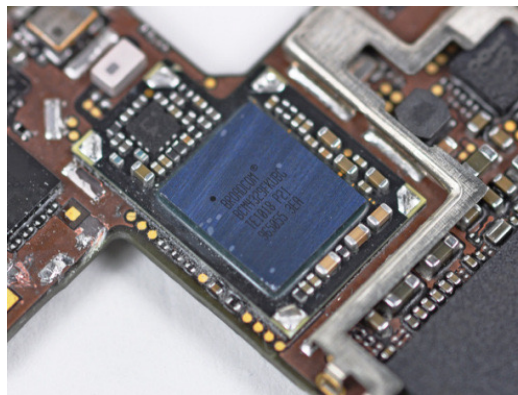
Does J.D. Power need a larger sample in 2024 than in 2009?

What's the point?

- Take any sort of data . . .
 - HALT scores
 - Sales data
 - Coin flips
 - Survey data
- By looking at means (and checking conditions), you can turn all of these data that have nothing to do with each other into the same sort of problem
- And, it's a problem that we **understand**
 - Convert any normal into a **standard normal** and work with the Z table in your book

When is a process out of control?

- A manufacturer of GPS chips selects samples for highly accelerated life testing (HALT)
- How should managers monitor these tests to ensure proper functioning of the production process?



Control limits

Thresholds that determine whether a manager should act or not

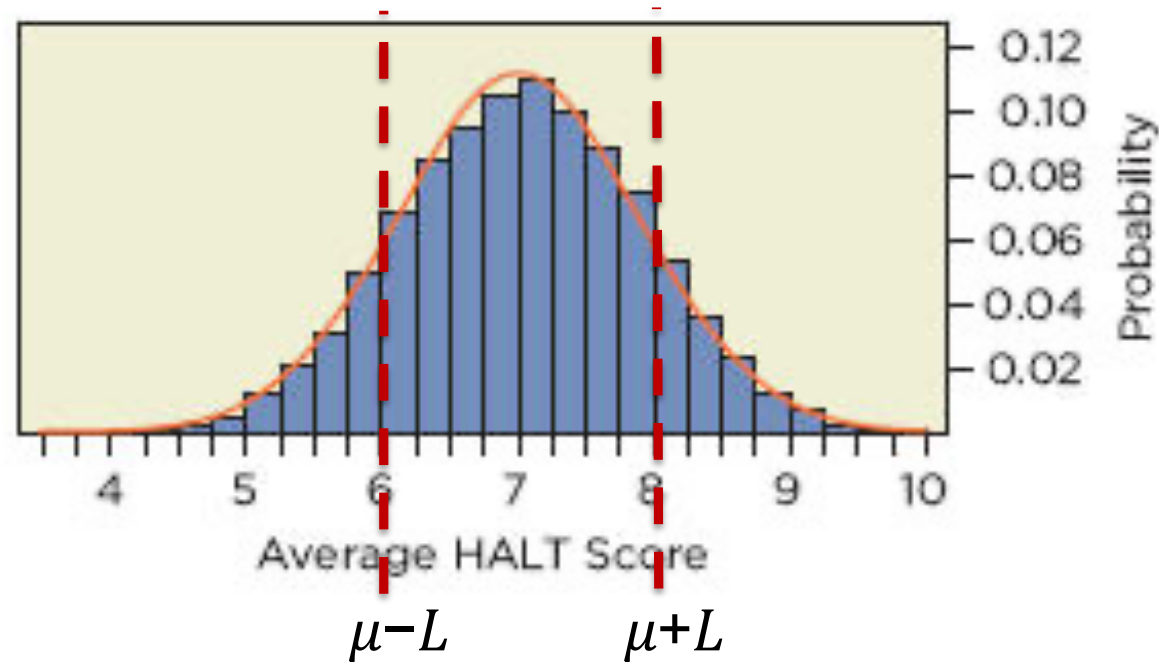
For example, is a process is out of control, or should it be allowed to continue?

- Symmetric interval denoted as:

$$\mu - L \leq \bar{X} \leq \mu + L$$

- Upper Control Limit (UCL) is $\mu + L$
- Lower Control Limit (LCL) is $\mu - L$
- What does a larger L mean for Type I and Type II errors?

Control limits

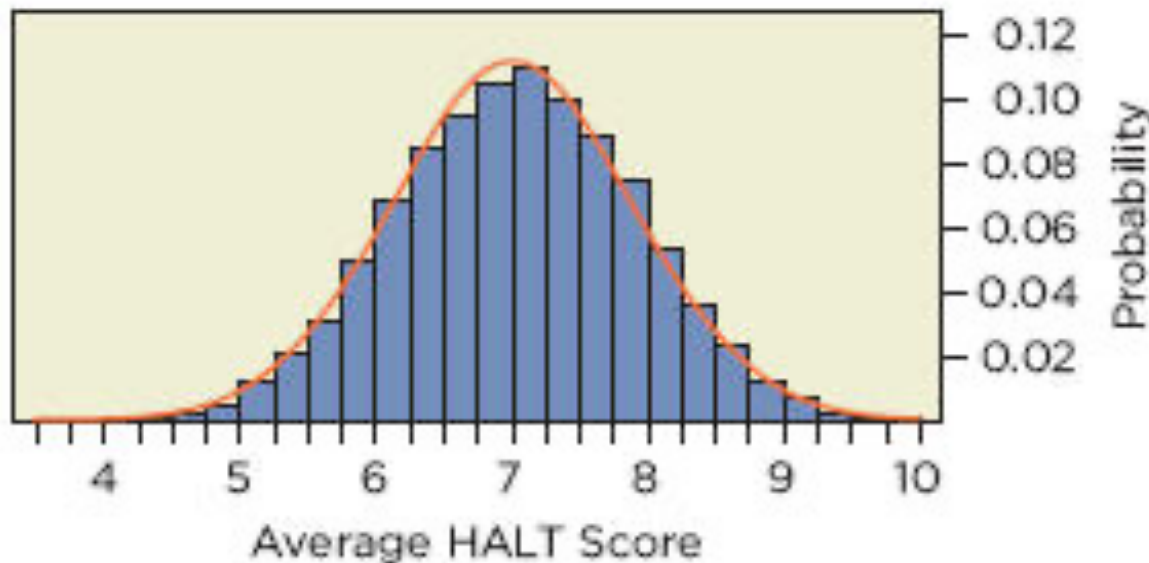


Decision Rule:

1. If your sample average falls outside the bars, you stop the machines.
2. If it falls inside, you keep them running.

Control Limits and Type 1 Error

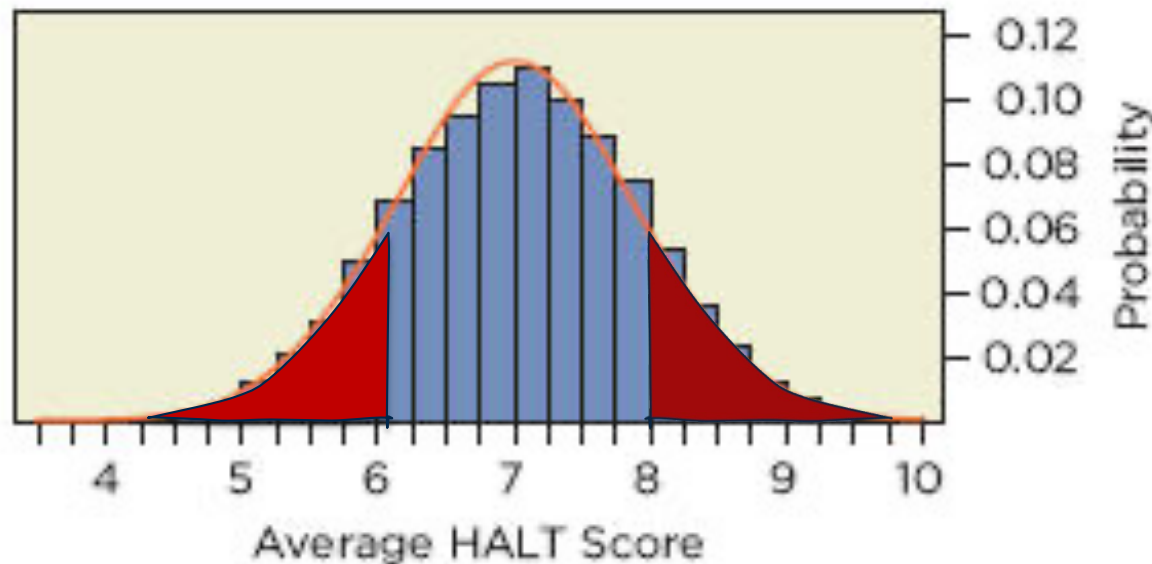
Suppose $L = 1$. If production is shut down when the mean HALT score is less than 6 or more than 8, what is the chance of a Type I error?



Sampling distribution of $\bar{X}$ when everything is OK

Control Limits and Type 1 Error

Suppose $L = 1$. If production is shut down when the mean HALT score is less than 6 or more than 8, what is the chance of a Type I error?



Everything is OK,
but your sample
falls in the red by
chance & you stop.
False positive.

Control Limits and Type 1 Error

If production is shut down when the mean HALT score is less than 6 or more than 8?

- What is the chance of Type I error?

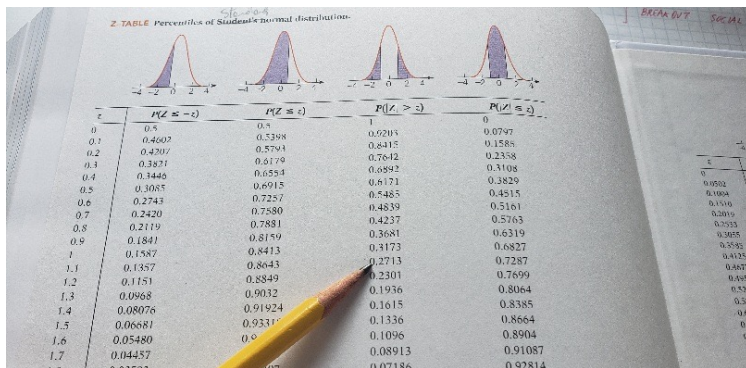
$$\begin{aligned}P(\bar{X} < 6 \text{ or } \bar{X} > 8) &= 1 - P(6 \leq \bar{X} \leq 8) \\&= 1 - P\left(\frac{6-7}{0.89} \leq \frac{\bar{X}-7}{0.89} \leq \frac{8-7}{0.89}\right) \\&= 1 - P(-1.1 \leq Z \leq 1.1) \\&= 0.27\end{aligned}$$

Control Limits and Type 1 Error

Suppose $L = 1$. If production is shut down when the mean HALT score is less than 6 or more than 8, what is the chance of a Type I error?

$$\begin{aligned}
 P(\bar{X} < 6 \text{ or } \bar{X} > 8) &= 1 - P(6 \leq \bar{X} \leq 8) \\
 &= 1 - P\left(\frac{6-7}{0.89} \leq \frac{\bar{X}-7}{0.89} \leq \frac{8-7}{0.89}\right) \\
 &= 1 - P(-1.1 \leq Z \leq 1.1) \\
 &= 0.27
 \end{aligned}$$

Using z-scores allows to focus on Standard Normal table



The image shows a Z-table with four columns: $P(Z \leq z)$, $P(Z \leq z)$, $P(Z > z)$, and $P(Z \leq z)$. The rows are labeled with z-scores from 0 to 1.7. A yellow pencil is pointing to the value 0.27 in the table, which corresponds to the probability $P(-1.1 \leq Z \leq 1.1)$.

z	$P(Z \leq z)$	$P(Z \leq z)$	$P(Z > z)$	$P(Z \leq z)$
0	0.5	0.5	0.5	0.5
0.1	0.5400	0.5398	0.4599	0.4597
0.2	0.5793	0.5791	0.4207	0.4205
0.3	0.6179	0.6177	0.3821	0.3819
0.4	0.6554	0.6552	0.3446	0.3444
0.5	0.6915	0.6913	0.3085	0.3083
0.6	0.7274	0.7272	0.2726	0.2724
0.7	0.7643	0.7641	0.2357	0.2355
0.8	0.8023	0.8021	0.1977	0.1975
0.9	0.8413	0.8411	0.1587	0.1585
1.0	0.8790	0.8788	0.1210	0.1208
1.1	0.9154	0.9152	0.0846	0.0844
1.2	0.9505	0.9503	0.0495	0.0493
1.3	0.9848	0.9846	0.0152	0.0150
1.4	0.9980	0.9979	0.0020	0.0019
1.5	0.9998	0.9997	0.0001	0.0001
1.6	0.9999	0.9999	0.0000	0.0000
1.7	0.9999	0.9999	0.0000	0.0000

Excel or Claude

$$\begin{aligned} P(\bar{X} < 6 \text{ or } \bar{X} > 8) &= \\ &= \text{NORMDIST}(6,7,.89,\text{TRUE}) + [1-\text{NORMDIST}(8,7,.89,\text{TRUE})] \\ &= 0.27 \end{aligned}$$

what is $\Pr(\bar{x} < 6 \text{ or } \bar{x} > 8)$ with $N(7, 0.89)$

Thought for 5s >

By symmetry, this is $P(|\bar{x} - 7| > 1)$.

If 0.89 is the standard deviation (standard error):

$z = 1/0.89 \approx 1.124$, so $P = 2 \times P(Z > 1.124) = 2 \times 0.1306 \approx \mathbf{0.261}$

If 0.89 is the variance ($sd = \sqrt{0.89} \approx 0.943$):

$z = 1/0.943 \approx 1.060$, so $P = 2 \times P(Z > 1.060) = 2 \times 0.1446 \approx \mathbf{0.289}$

Since you wrote it as $N(7, 0.89)$ with $\bar{x}$, I'd guess 0.89 is the standard error of the mean, making 0.261 the answer — but worth double-checking which convention the problem uses.

Control Limits

Balancing Type I and Type II Errors

- Wide control limits reduce the chance for a Type I error
- Narrow control limits reduce the chance for a Type II error
- Cannot simultaneously reduce the chances of both
- The probability of a Type I error is typically set at 5% or 1%.

Control Limits

Probability of a Type I Error (α)	z-Score Cutoff ($z_{\alpha/2}$)
.05	1.96
.01	2.58
.0027	3.00

The control limits for a sample of n measurements from a process with mean μ and standard deviation σ are,

$$\mu \pm z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}.$$

Repeated testing

- In settings in which testing is repeated over and over, companies often use wide control limits (e.g., $\alpha = .0027$).
- Otherwise, with consecutive tests, the probability of Type I errors can become unacceptably large.
 - For example, a 5% probability of a Type I error in one test implies a 40% probability over 10 consecutive tests.
 - Why? $1 - .95^{10} \sim 0.40$
- Repeated testing: Companies often willing to accept higher probability of Type II error for given test because they will have multiple opportunities to detect any problems.

Takeways

- Think hard about which attribute of the process to monitor.
- Do not concentrate on one error while ignoring the other.
- Do not assume that the process has failed if a value appears outside the control limits.
- Avoid confusing Type I and Type II errors.

Punchline

Making decisions with incomplete information is hard

- Even the best decision-making process will produce some mistakes

Normality of sample mean gives us a precise measure of how likely different outcomes are

- Allowing us to set action thresholds that specify risk of Type I errors

Bonus: Sampling Distribution of the Mean

What is it about rural counties and kidney cancer?

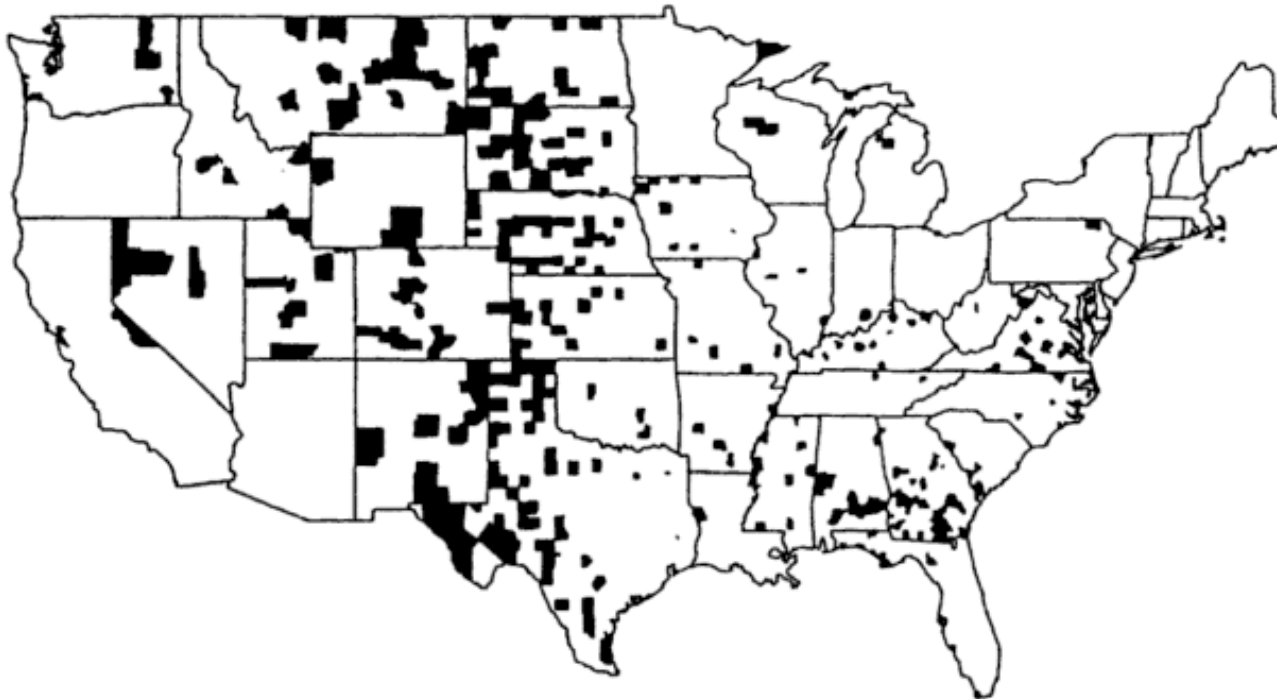


Figure 1.1.
Lowest kidney cancer death rates. The counties of the United States with the lowest 10% age-standardized death rates for cancer of kidney/urethra for U.S. males, 1980–1989 (from Gelman and Nolan, 2002, p. 15, reprinted with permission).

Bonus: Sampling Distribution of the Mean

What is it about rural counties and kidney cancer?

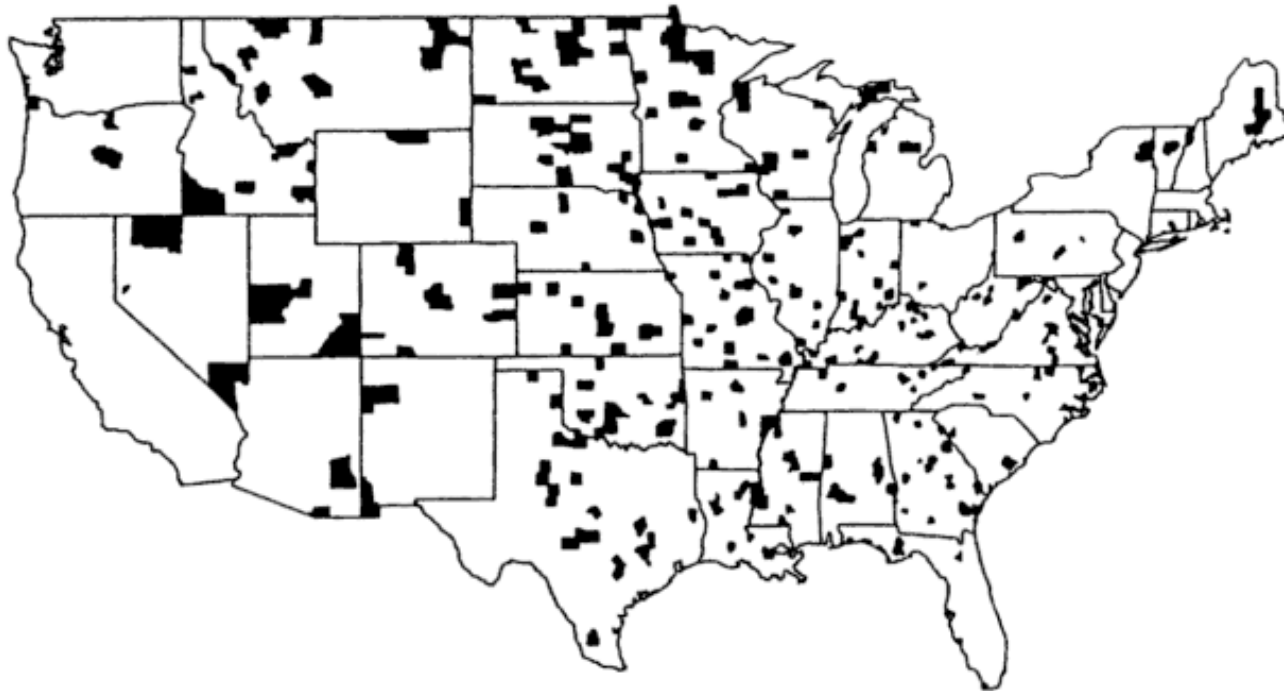


Figure 1.2.
Highest kidney cancer death rates. The counties of the United States with the highest 10% age-standardized death rates for cancer of kidney/urethra for U.S. males, 1980–1989 [from Gelman and Nolan, 2002, p. 14, reprinted with permission].

Bonus: Sampling Distribution of the Mean

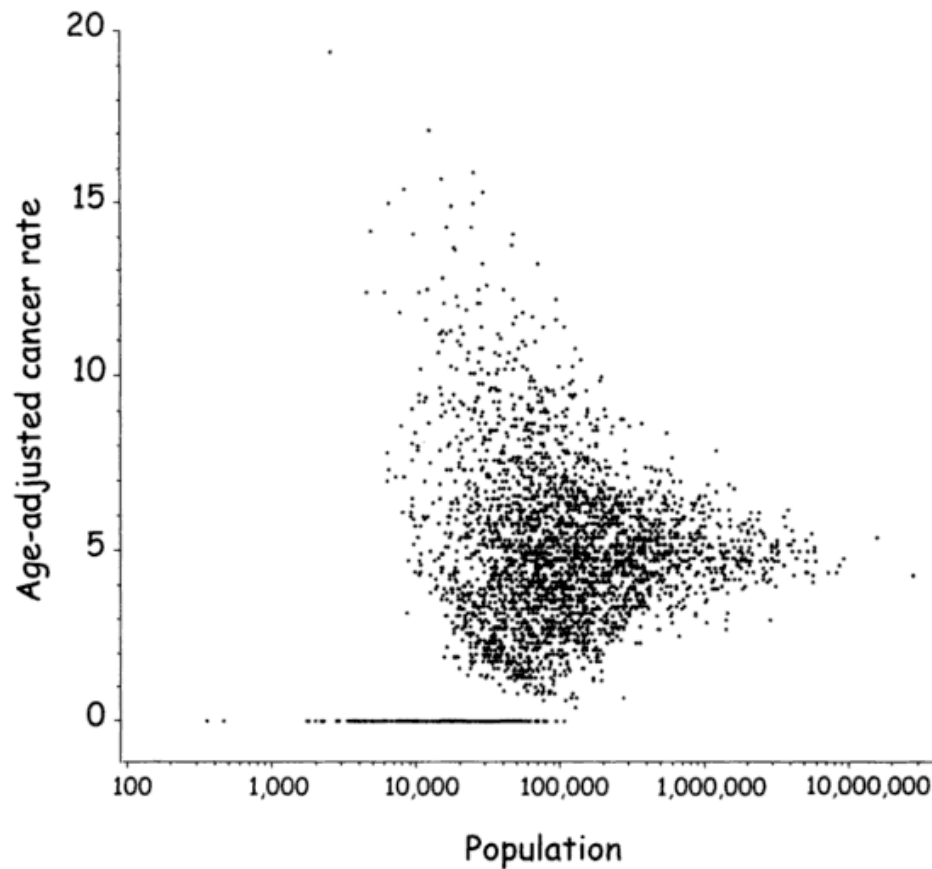


Figure 1.3.
Age-adjusted kidney cancer rates for all U.S. counties in 1980–1989 shown as a function of the log of the county population.

Gambling on a personnel decision?

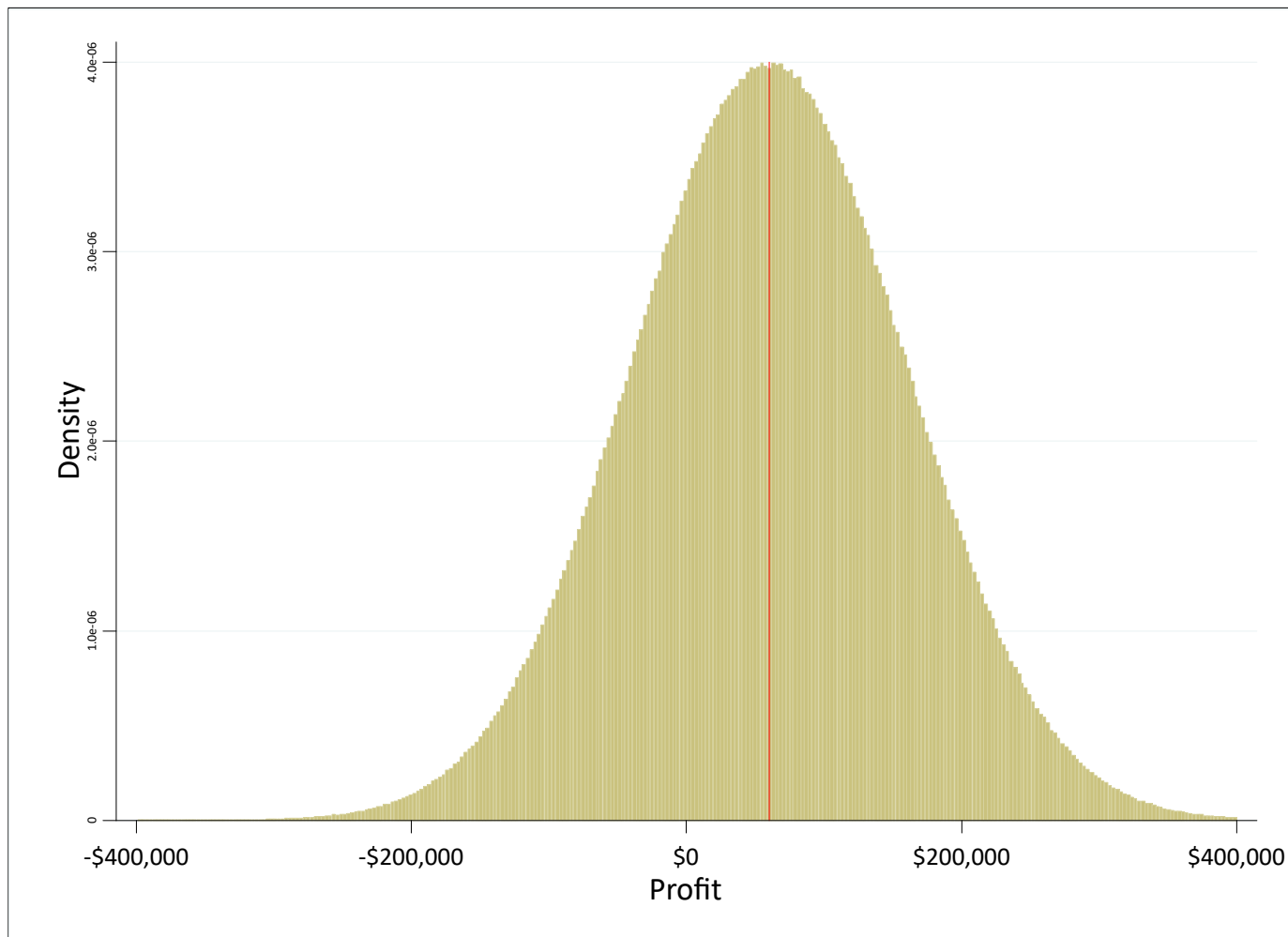
Meet “Jimmy Quickfingers”

- Blackjack Dealer at Caesars Palace
- Jimmy’s table has lost an average of \$12,000 per night over the last month
- Is Jimmy stealing from the casino? Should he be fired?

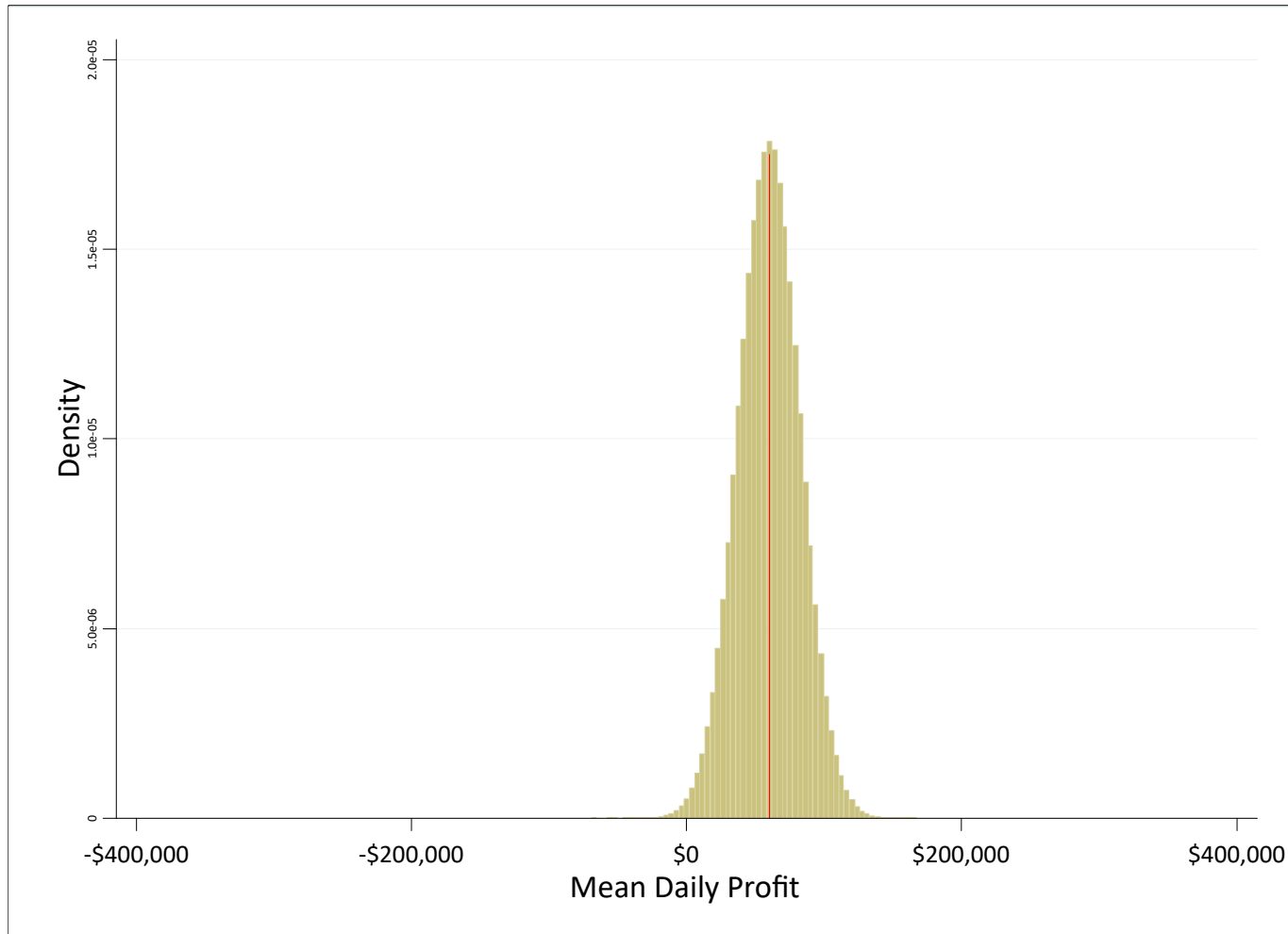


Distribution of Daily Profit

When dealers are honest, tables earn on average a nightly profit of \$60,000 with a standard deviation of \$100,000



Distribution of Average Daily Profit By Month



*Suppose dealers work 20 days per month . . .
Describe the distribution of the sample average*

Distribution of **average** daily profit in a month

- The standard deviation of daily profit is \$100,000 and there are 20 working days per month.
- Therefore, the standard error for average daily profit per month is,

$$\frac{\sigma}{\sqrt{n}} = \frac{\$100,000}{\sqrt{20}} \cong \$22,360.$$

- Thus average daily profit in a month has a distribution that is approximately normal, with mean \$60,000 and standard error \$22,360.

Probability of Losing \$12,000 Per Day

What is the probability that an honest dealer loses \$12,000 or more per day over the course of a month?

$$P(\bar{X} \leq -\$12,000) =$$

$$P\left(\frac{\bar{X} - \$60,000}{\$22,360} \leq \frac{-\$12,000 - \$60,000}{\$22,360}\right) =$$

$$P(Z \leq -3.21)$$

Using the Z-table this corresponds to .0007. This is less than 1 in a 1000.

- So do you fire Jimmy? Are you sure you're right?

Calculating the probability of a Type I error

- Suppose we followed the rule of firing an employee anytime they had average daily profit for a month that was below \$0 or above \$120,000.
- If that was our rule, what is the probability of a Type I error?
- From a business perspective, is this an acceptable probability?
- Suppose you used wider limits.
 - How would that change the probability of a Type I error?
 - How would that change the probability of a Type II error?

Calculating the probability of a Type I error

- The question becomes: “How likely are observations below 0 and above 120,000 given the distribution of sample averages?”
- We don’t have a table for our distribution, so you need to convert values into the standard normal
 - I like to think about this in terms of distances measured in units of standard errors
 - How many standard errors away from the mean is 0?
 - How many standard errors away from the mean is 120,000?

Calculating the probability of a Type I error

Suppose we followed the rule of firing an employee anytime they had average daily profit for a month that was below \$0 or above \$120,000.

If that was our rule, what is the probability of a Type I error?

$$\begin{aligned} P(\bar{X} < \$0 \text{ or } \bar{X} > \$120,000) &= 1 - P(\$0 \leq \bar{X} \leq \$120,000) \\ &= 1 - P\left(\frac{\$0 - \$60,000}{\$22,360} \leq \frac{\bar{X} - \$60,000}{\$22,360} \leq \frac{\$120,000 - \$60,000}{\$22,360}\right) \\ &= 1 - P(-2.67 \leq Z \leq 2.67) \end{aligned}$$

Using the Z-table this corresponds to about .008; slightly less than 1 percent.